EE2004: Digital Signal Processing

Tutorial 10

(April 16, 2026)

1. Find the Nyquist sampling rates for the following signals: (a) $\text{sinc}(20t)$, (b) $4\,\text{sinc}^2(100t)$, (c) $8\,\sin(50\pi t)$, (d) $4\,\sin(30\pi t) + 3\,\cos(70\pi t)$, (e) $\text{rect}(300t)$.

2. Let $x_c(t) = \cos(4000\pi t)$.
   (a) If the signal is sampled every $T = 1/6000$ s, will there be any aliasing? Sketch $X_s(F)$.
   (b) If the signal is sampled every $T = 1/1500$ s, will there be any aliasing? Sketch $X_s(F)$.

3. The signal $x_c(t) = \cos(2\pi F_0 t + \phi)$ is sampled at F_s Hz where $F_0 = 8$ kHz and $F_s = 12$ kHz, giving rise to $x[n] = \cos(\omega_0 n + \theta)$. It is given that $0 < \omega_0 < \pi$. Determine the value of (i) ω_0 and (ii) θ in terms of ϕ.

4. The sequence $\phi[n] = \cos\left(\frac{\pi}{4}n\right)$, was obtained by sampling a continuous-time signal $\phi(t) = \cos(\omega_0 t)$, at a sampling rate of 1000 samples/s. What are two possible positive values of ω_0 that could have resulted in the sequence $\phi[n]$?

5. The continuous-time signal $\phi(t) = 4\sin(20\pi t) - 5\cos(24\pi t) + 3\sin(120\pi t) + 2\cos(176\pi t)$ is sampled at $50Hz$ rate, generating the sequence $\phi[n]$. Determine the exact expression of $\phi[n]$.

6. Let $x(t) = \sum_{-10}^{10} jk\, e^{j16\pi kt}$. Find Nyquist sampling rate.

7. The spectrum $X(F) \in \mathbb{R}$ of a certain continuous-time signal is triangular in shape centred at $F = 0$ with $X(0) = 1$ and $X(F) = 0$ for $|F| > F_c$. The underlying $x(t)$ is sampled at F_s Hz, with $x_s(t)$ being its associated impulse-train sampled signal and $x[n]$ being the corresponding discrete-time sequence. Find $X_s(F)$, $X_s(j\Omega)$, $X(e^{j\omega})$, and $X(e^{j2\pi F})$ when (a) $2F_c < F_s$, and (b) $2F_c > F_s$.

8. Let $x(t)$ be a continuous time signal having Nyquist frequency Ω_c. Find Nyquist frequencies of the following signals in terms of Ω_c : (a) $x_1(t) = x^2(t)$, (b) $x_2(t) = x(t/5)$, (c) $x_3(t) = x(4t)$, (d) $x_4(t) = \int_{-\infty}^{\infty} x(t-\tau)x(\tau)\,d\tau$, and (e) $x_5(t) = \dfrac{dx(t)}{dt}$.

9. The continuous-time signal $x_c(t) = \sin(20\pi t) + \cos(40\pi t)$ is sampled with a sampling period T to obtain the discrete-time signal $x[n] = \sin\left(\dfrac{\pi n}{5}\right) + \cos\left(\dfrac{2\pi n}{5}\right)$.
   (a) Determine a choice for T consistent with this information.
   (b) If your choice of T is unique, explain why. If not, specify another choice of T consistent with this information.

10. **Computer assignment** The ideal reconstruction filter implements the following computation:

$$x_c(t) = \sum_{n=-\infty}^{\infty} x[n] \frac{\sin[\pi(t-nT)/T]}{\pi(t-nT)/T}$$

which is non-realizable because it requires an infinite number of samples. Instead consider,

$$\hat{x}_c(t) = \sum_{n=0}^{N} x[n] \frac{\sin[\pi(t-nT)/T]}{\pi(t-nT)/T}$$

which uses only $(N+1)$ terms. We will study the resulting error $e(t) = x_c(t) - \hat{x}_c(t)$ in the interval $0 \le t \le NT$. Most observations can be made with $3 \le N \le 20$.

(a) Develop a MATLAB function `[xhat,t] = DAC(x,N,T)` that implements the finite sum over the interval $0 \le t \le NT$ given the samples $x[n]$ over $0 \le n \le N$ and the sampling interval T. To obtain a smooth graph for $\hat{x}_c(t)$, choose at least 20 points per sampling interval and provide the resulting time vector `t` as the second output parameter.

(b) Let $x_c(t) = 1$. Plot $x_c(t)$, $\hat{x}_c(t)$, and $e(t)$ for $T = 1$ and $N = 5$, 10, 15, and 20. Explain the shapes of the plotted signals. How does the error behave as a function of t?

(c) Let the "size" of the error be defined as the maximum value of $|e(t)|$ in the middle third of the plotted time interval. Determine and plot the size of error as a function of N for $3 \le N \le 20$. Comment on your results.

(d) Repeat the previous two parts for $x_c(t) = \cos(2\pi Ft)$ with $F = 1$ and $T = 0.1$. Compare your results with the $x_c(t) = 1$ case.

(e) Repeat part (d) for various values of F and $T = 1/F_s$. Is the size of the error dependent on the normalized frequency $f = F/F_s$?

(f) Now suppose $x_c(t) = \cos(2\pi Ft + \theta)$ with $F = 1$ and $T = 0.1$, and consider various values of θ. Do the results depend on θ?

11. **Computer assignment** Consider the first-order system described by $y[n] = 0.8\,y[n-1] + 0.2\,x[n]$. It is excited by by the *linear FM signal* given by

$$x[n] = \cos\left[\pi\left(\frac{B}{F_s N}\right)n^2\right] \qquad 0 \le n \le N$$

where $B = 10\,\text{Hz}$, $F_s = 100\,\text{Hz}$, and $N = 1000$.

(a) Determine and plot $|H(e^{j2\pi F})|$ over $0 \le F \le B\,\text{Hz}$.

(b) Plot $x[n] = x(nT_s)$ over $0 \le t \le \tau\,\text{sec}$.

(c) Process the input $x[n]$ through the system using MATLAB to obtain $y[n] = y(nT_s)$ and plot it over $0 \le t \le \tau\,\text{sec}$. Verify that the amplitude of $x[n]$ is attenuated according to the frequency response $H(e^{j2\pi F})$.

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